Laure Moinat

Curriculum Vitae

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RESEARCH FOCUS

I study how nonlinear interactions within the Earth system give rise to multiple climate states, tipping behavior, and state-dependent climate responses. My work combines coupled climate modeling, nonlinear dynamics, and paleoclimate to understand the emergence and stability of Earth's multiple equilibrium states and the transitions between them.

EDUCATION

2023 - (expected June 2027): Ph.D in physics - Nonlinear climate dynamics, University of Geneva, Switzerland

2021 - 2023: Master of Science in physics, specialization in applied physics, University of Geneva, Switzerland

2018 - 2021: Bachelor of Science in physics, University of Geneva, Switzerland

2018: Bilingual International Baccalaureate (French/English C2 level), International School of Geneva, Switzerland
Prize: Merit and distinction for academic achievements

PROFESSIONAL EXPERIENCE

2023 - ongoing: PhD researcher and graduate teaching assistant, Physics section, University of Geneva, Switzerland
PhD subject: 'Climate Multistability and Transitions: From Earth's Past to Future' supervised by Dr. Maura Brunetti, Dr. Christian Vérard and Prof. Jérôme Kasparian

Research contributions:

1. Climate model development: Lead developer of *BIG-MITgcm*, a coupled ocean–atmosphere–ice-sheet–vegetation–hydrology framework designed to explore equilibrium climate states and long-timescale Earth-system feedbacks.
2. Climate multistability and nonlinear Earth-system dynamics: Identified and characterized multiple equilibrium climate states under present-day boundary conditions and investigated their dependence on atmospheric CO_2
3. Tipping detection and critical transitions: Developed climate network approaches to identify early warning signals and characterize unstable states associated with tipping in the AMOC and other climate components.

2022 - ongoing: Extracurricular teacher, Collège du Léman, Switzerland
Provide individualized instruction in mathematics, physics, and chemistry to secondary-school students following the Swiss Maturity, French Baccalaureate, International Baccalaureate, US High School Diploma, and Advanced Placement curricula in both English and French.

2021 - 2023 : Master student and undergraduate teaching assistant, Physics section, University of Geneva, Switzerland
Master thesis: "Climate change velocity computation applied to ecology" supervised by Prof. Jérôme Kasparian

GRANTS AND AWARDS

2026: **Subside Tremplin award, University of Geneva - 30000 CHF**
Competitive early-career award supporting academic development and future research independence.
Interview for the University: https://mediaserver.unige.ch/play/307686

2025: Young Researcher and Innovator Grant, COST action - 1400 €
Global tipping point conference: Detection of global-scale tipping using climate networks

2025: European Geosciences Union Outstanding Student and PhD candidate (OSPP) award - 2300€
EGU GA: Development of a biogeodynamical tool for exploratory paleoclimate modeling
Awarded poster and small biography: https://www.egu.eu/awards-medals/ospp-award/2025/laure-moinat/

2025: Young Researcher and Innovator Grant, COST action - 1000€
EGU GA: A biogeodynamical tool for exploratory paleoclimate modelling

2024: ProClim Best poster award, Swiss Academy of Sciences - 1000 CHF
Swiss Global Change Day: Investigating Early Warning Signals using Complex Networks
Awarded poster: https://proclim.scnat.ch/fr/24thswissglobalchangeday

PUBLICATIONS

Journal Articles

*Share first authorship, +Under review or in preparation

Lead author

1. **Moinat, L.**, Kasparian, J., and Brunetti, M. (2024). Tipping detection using climate networks. *Chaos*, 34(12), 123161. https://doi.org/10.1063/5.0230848 ***(Editor's pick)***

2. **Moinat L.**, Gaponenko I., Goyette S. et al. Comparing ecological relevance of climate velocity indices *Sci. Rep.* 16, 8797 (2026). https://doi.org/10.1038/s41598-025-32377-0

3. **Moinat, L.**, Franziskakis, F., Vérard , C., Goldberg, D., and Brunetti, M.: Biogeodynamics-Icesheet-Geneva-MITgcm (BIG-MITgcm, v1.0): a simulation tool for exploring climate states with a representation of global ice sheets, *Geosci. Model Dev.*, 19, 4357–4384, https://doi.org/10.5194/gmd-19-4357-2026, 2026. ***(Invited contribution)***

4. **Moinat, L.**, Börner, R., Lucarini, V., Brunetti, M., & Dijkstra, H. A. (2026). Climate network characterization of the Atlantic Meridional Overturning Circulation edge state. *Phys. Rev. Research*, 8, 033268. https://doi.org/10.1103/q9bd-4h4j

5. +*Sternai P., *__Moinat L.__, Meroni N. A., Pasquero C., Brunetti M. and Vérard C: Earth's mantle dynamics shape climate and the Earth system *(under review at Science Advances)*

6. +__Moinat L.__, Marshall J., and Brunetti M: Competing climates of the present-day Earth: from snowball to hothouse *(to be submitted Science Advances)*

7. +__Moinat L.__, Vérard C., Goldberg D. N., Kasparian J., Gerya T., Marshall J. and Brunetti M: Effect of the Milankovitch cycles on climate multistability for the last 1 Myr *(in preparation)*

8. +__Moinat L.__, Scott J., Marshall J., Ferreira D., Campin J-M., and Brunetti M: Northern Pacific sea ice extent controls the stability of colder climate states *(in preparation)*

Collaborative projects

9. +Aben J., Travis MJ. J., Ascensao F., Bessa-Gomes C., van Dongen S., Duval G., **Moinat L.**, Kasparian J. and Clouteau D: Harnassing transport networks to enhance climate connectivity for biodiversity *(under review at Global Change Biology)*

10. +Tsigaridis K., Sohl L., Fauchez T., Bhatnagar S., Brunetti M., Chaverot G., Gomez-Leal I., Haqq-Misra J., Harman C., Iro N., Jaziri A., Kofman V., Lambert H. F., MacDonald E., **Moinat L.**, Paradise A., Sergeev D., Simonetti P., Skinner J., Thomson I. S., Turbet M., Williams A. D. and Wolf E.: The Comparison of Reference Exoplanet Models of Earth (CREME) protocol version 1.0: A CUISINES Model Intercomparison Project *(under review at Planetary Science Journal)*

Co-supervised Master thesis

1. LUCET, Thomas Vitalis. Modelling climate tipping-point cascade: the interlinked evolution of ocean circulation and Greenland ice sheet. Master, 2026. https://archive-ouverte.unige.ch/unige:193882

PRESENTATIONS and COLLABORATIONS

Conference presentations

15 conference presentations, including **7 oral presentations** at EGU General Assembly, AGU Annual Meeting, Ocean Sciences Meeting, Swiss Ocean Day, and international workshops.

17. **Moinat, L.**, Marshall, J., and Brunetti, M. Multiple climate attractors under the present day continental configuration: mechanisms and implications, *AGU Annual Meeting 2026 (December 2026)*, Poster

16. **Moinat, L.**, Börner, R, Lucarini, V., Brunetti, M. and Dijkstra H. A. Climate networks unveil the role of the AMOC edge state during destabilization, *AGU Annual Meeting 2026 (December 2026)*, **Geoburst (Talk + Poster)**

15. **Moinat, L.**, Marshall, J., and Brunetti, M. Multiple climate attractors under the present day continental configuration: mechanisms and implications, *Swiss Geoscience Meeting (November 2026)*, Poster

14. **Moinat, L.**, Vérard, C, Goldberg, D. N., Kasparian, J., Gerya, T., Marshall, J., and Brunetti, M. Effect of the Milankovitch cycles on climate multistability for the last 1 Myr, *EUROBIG General Assembly (July 2026)*, Poster

13. **Moinat, L.**, Börner, R, Lucarini, V., Brunetti, M. and Dijkstra H. A. Climate networks unveil the role of the AMOC edge state during destabilization, *Swiss Ocean Day (June 2026)*, **Talk**

12. **Moinat, L.**, Vérard, C, Goldberg, D. N., Kasparian, J., Gerya, T., Marshall, J., and Brunetti, M. Effect of the Milankovitch cycles on climate multistability for the last 1 Myr, *EGU General Assembly (May 2026)*, **Talk**

11. **Moinat, L.**, Börner, R, Lucarini, V., Brunetti, M. and Dijkstra H. A. Spatial signature of the AMOC edge state using climate networks, *Ocean Science Meeting (March 2026)*, **Talk**

10. **Moinat, L.**, Vérard, C, Goldberg, D. N., Kasparian, J., Gerya, T., and Brunetti, M. Climate multistability and its robustness against the Milankovitch cycles for the last 1 Myr, *AGU General Assembly (December 2025)*, Poster

9. **Moinat, L.**, Lucet, T, Kasparian, J., and Brunetti, M. Modelling tipping cascades: the interlinked evolution of the Amazon rainforest, ocean circulation and the Greenland ice sheet, *AGU General Assembly (December 2025)*, Poster

8. **Moinat, L.**, Kasparian, J, and Brunetti, M. Detection of global-scale tipping using climate networks, *Global tipping points: accelerating action to ensure a socially just transformation (July 2025)*, Poster.
Received the Young Researcher and Innovator Grant

7. **Moinat, L.**, Franzizkakis, F., Vérard, C., Goldberg, D. N., and Brunetti, M. Development of a new biogeodynamical tool for exploratory climate modelling, *EUROBIG General Assembly (June 2025)*, Poster

6. **Moinat, L.**, Franzizkakis, F., Vérard, C., Goldberg, D. N., and Brunetti, M. Development of a biogeodynamical tool for exploratory paleoclimate modelling, *EGU General Assembly (April/May 2025)*, Poster.
Received the OSPP award (Outsanding Student and PhD candidate) + Young Researcher and Innovator Grant

5. **Moinat, L.**, Franzizkakis, F., Vérard, C., Goldberg, D. N., and Brunetti, M. Development of a new biogeodynamical tool for exoclimate modeling, *Swiss Habitability and Origin of Life (March 2025)*, Poster

4. **Moinat, L.**, Kasparian, J., and Brunetti, M. Detection of global-scale tipping using climate networks, *Metastability and Tipping in Complex Systems (September 2024)*, **Talk**

3. **Moinat, L.**, Kasparian, J., and Brunetti, M. Investigating Early Warning Signals in Climate Simulations using Complex Networks, *EGU General Assembly (April 2024)*, Poster

2. **Moinat, L.**, Kasparian, J., and Brunetti, M. Investigating early warning signals using climate networks, *24th Swiss Global Change Day (April 2024)*, Poster.
Received the ProClim Best poster award

1. **Moinat, L.**, Kasparian, J., and Brunetti, M. Signature of Climate Dynamics using Complex Networks, *Swiss Habitability and Origin of Life (March 2024)*, **Talk**

Collaborative projects

- **2023 - ongoing: Base to top Earth SNF/Sinergia project**, University of Geneva and ETH Zurich, Switzerland
Climate-modeling lead for simulations of key Phanerozoic climate transitions; integrating climate, paleogeography, and geodynamic modeling.
- **2023 - ongoing: Centre pour la vie dans l'Univers (CVU)**, University of Geneva, Switzerland
Associated researcher related to the Base to top Earth project.
- **2024 - ongoing: pan-EUROpean BIoGeodynamics network (EUROBIG)**, COST action
Contributing to model-development activities within WG3 and science/outreach activities within WG6.
- **2024 - ongoing: TIPMIP: TIpping Points Modeling Inter-comparison Project**, PIK Postdam, Germany
Developed network-based approaches for identifying early-warning signals at local and global scales.
- **2023- ongoing: CUISINES: CREME group (exoplanet model inter-comparison)**, Columbia University, USA
Comparing *BIG-MITgcm* with other models and its potential application to exoplanets.

TEACHING, SUPERVISION and EDUCATIONAL OUTREACH

Co-supervision of master students:

2023 - ongoing: Master of Science in environmental science, University of Geneva, Switzerland

- Kira Kalinichenko: *Evaluation of vegetation tipping points using climate networks*, ongoing
- Byeongseok Kang: *Impact of paleogeographic reconstructions on biomes in the Permian-Triassic*, ongoing
- Thomas Lucet: *Modeling tipping cascades: the interlinked evolution of the Amazon rainforest, ocean circulation and the Greenland ice sheet* (Grade: **6/6** and **1 publication in preparation**)

Teaching experience:

2021 - ongoing: Graduate/undergraduate teaching assistant, Physics section, University of Geneva, Switzerland

- Introduction to climatology and meteorology (M.Sc in Environemental science): Guest lecturer covering an introduction to ocean overturning circulation and tipping elements in the climate system (taught in English)
- Nonlinear Systems (M.Sc. in Physics): Covering nonlinear dynamics and chaos theory in the climate system. Responsible of the exercises session during the semester: I redesigned the assessment process to promote student interaction and active learning. Co-responsible of the evaluation during the oral and written examinations (taught in English)
- General Physics A (B.Sc in Medicine): General physics course. I contributed to the creation of the exercises sheet, the exam, and explaining the exercises (taught in French)

- Mathematical methods for Physicists I (B.Sc in Physics): Applied mathematics course. Led exercise sessions and supported students in solving problems (taught in French)

Educational Outreach:

- Dies Academicus: participated in a video highlighting six researchers from the University, shown during the official ceremony Dies Academicus in front of Geneva officials and broadcast on Léman Bleu TV.*(October 2026)*
  University video: https://www.youtube.com/watch?v=phyuT5-U0HU&t=237s
- MITgcm blog: communication about BIG-MITgcm on the MITgcm blog under *This Month's Featured Publication (June 2026)*
- Sketching Science: working with students from the HEAD (Haute Ecole d'Art et de Design) school, so that they produce an illustration to support a publication *(April 2026)*
- Ma Thèse en 180 secondes (MT180): participated in this competition that aims to communicate to a wider public my PhD research in 3 minutes, qualified for the Geneva final *(April 2026)*
  Portrait and presentation: https://www.unige.ch/mt180/lauremoinat
- Programme boussole: invited talk to present the research of the *Applied Physics* department to high school students who want to pursue their studies in physics visiting the university *(March 2025 and 2026)*
- La nuit de la Science: promote science and presenting the research done in the lab to a broader public in the city of Geneva *(June 2024)*

Service:

- Physics Section highlight: small biography and interview related to the Subside Tremplin award for the section website and newsletter *(May 2026)*
- Panel member (PhD student representative) for the CVU excellence chair professor nomination (responsible Prof. Emeline Bolmont, CVU director) *(May 2026)*
- Highlighted student: interviewed in order to promote the excellence in academic path after obtaining a diploma at the *International School of Geneva (February 2025)*
- Campus Coach: Coaching students at their arrival at the University (B.Sc and M.Sc in Physics) and during their first year *(September 2022 and 2023)*
- Welcome Days: Responsible for welcoming first year B.Sc Physics student by showing them the university and giving advice *(September 2021, 2022 and 2023)*
- Guiding video for the Science faculty: Participating in a video showing the *Faculty of Science* buildings and sharing my experience *(September 2021)*
- Member of the European Geosciences Union (EGU) and the American Geophysical Union (AGU)

WORKSHOPS and SCHOOLS

Workshops/Internal seminar:

- CVU day: flash talk on *Effect of the Milankovitch cycles on climate multistability for the last 1 Myr* , Observatory of Geneva, University of Geneva, Switzerland *(January 2026)*
- Physics department day: talk on *The dynamical structure of the Earth climate: implications for its trajectory under the current climate change*, Physics section, University of Geneva, Switzerland *(January 2026)*
- CVU day: flash talk on *Development of a biogeodynamical tool for exoclimate modeling*, Observatory of Geneva, University of Geneva, Switzerland *(January 2025)*
- Physics department day: flash talk and poster on *Identifying Tipping points in Climate Simulations using Complex Networks*, Campus Biotech, University of Geneva, Switzerland *(January 2024)*
- CVU day: flash talk on *Tipping points - spatial correlation*, Observatory of Geneva, University of Geneva, Switzerland *(November 2023)*

Schools:

- *Introduction to Geodynamic Modelling*, EUROBIG at the Polish Academy of Sciences, Poland *(September 2025)*
- *Young researcher training*, COST Academy at the COST association, Brussels, Belgium *(September 2025)*, **Invited**